Titanic Survival Prediction

A Machine Learning approach with Decision Tree Classifier

Luca Frittitta · Dataset: Titanic · Model: DecisionTreeClassifier (scikit-learn)

Objective

Goal

Predict whether a passenger survived the Titanic disaster using features like sex, age, class and port of embarkation.

Approach

Build and tune a Decision Tree Classifier, optimizing max_depth through a train/validation/test split strategy.

Outcome

Identify the best model hyperparameters and evaluate final performance on unseen test data.

Dataset Overview

891	5	38%	177
Passengers	Features	Survived	Missing Ages

Features Used

Feature	Type	Notes
Sex	Categorical	One-Hot Encoded (Sex_female / Sex_male)
Age	Numeric	177 NaN → filled with median
Pclass	Ordinal	1st, 2nd, 3rd class
Embarked	Categorical	One-Hot Encoded (C/Q/S)
Survived	Target	0 = No, 1 = Yes

Exploratory Data Analysis

Sex → Strongest predictor

Women: 74% survival
Men: 19% survival
Biggest feature gap in the dataset.

Passenger Class matters

1st class: 63% | 2nd: 47%
3rd class: 24% survival
Socioeconomic status was decisive.

Age vs Survived (new)

Younger passengers had slightly higher survival rates.
Distribution is right-skewed

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Port of Embarkation

Cherbourg (C): 55% survival
Queenstown (Q): 39%
Southampton (S): 34%

Data Preprocessing

01

Train / Val / Test Split

Split first — before any preprocessing — to avoid data leakage.
75% train+val → 25% test | then train+val: 75% train → 25% val
Final: ~501 train / ~167 val / ~223 test

02

Handle Missing Values

Age: median of training set (distribution is right skewed)

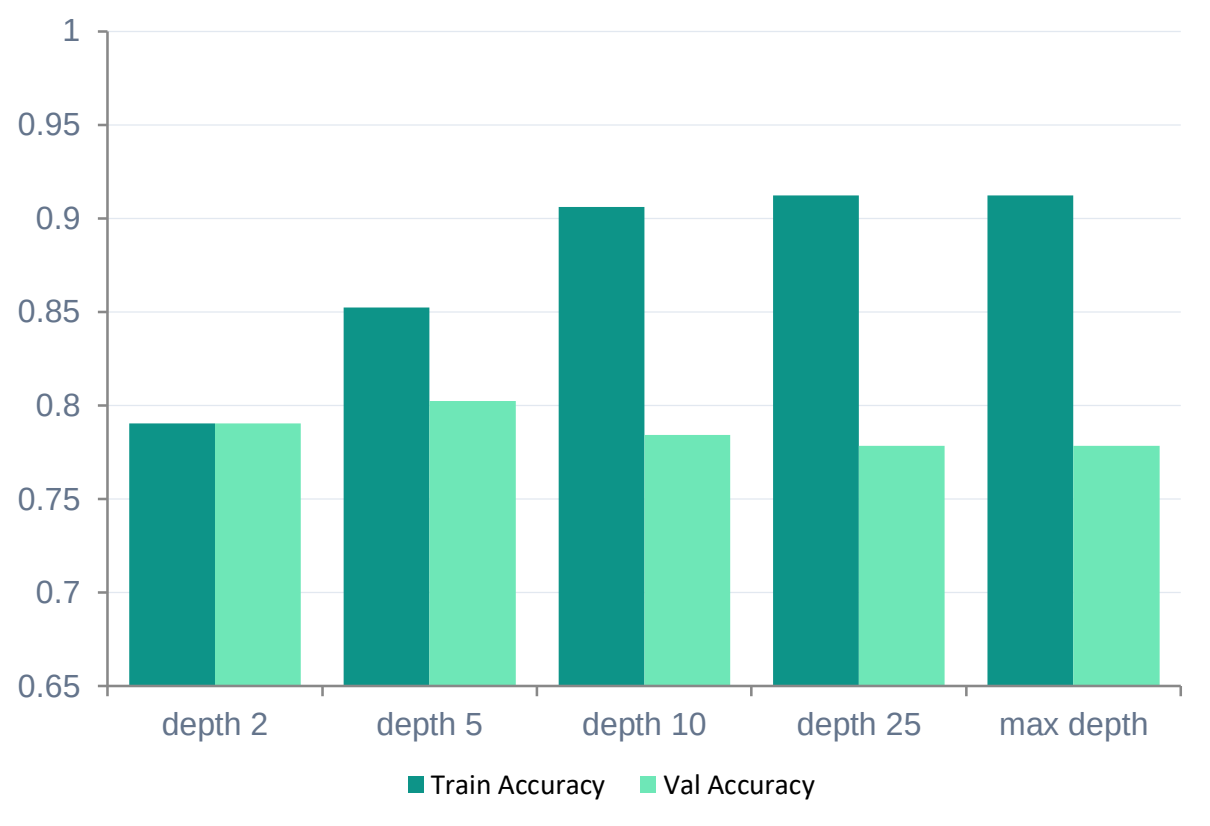
Embarked: mode of training set (most frequent = 'S')

03

Encoding Categorical Features

Sex → One-Hot Encoding (Sex_female / Sex_male)
Embarked → One-Hot Encoding (Embarked_C / Embarked_Q / Embarked_S)

Model Tuning — Finding the Best Depth



Best depth

max_depth = 5
Val accuracy: 80.24%

Depth 2

Val acc 79.0% — shallow tree, slightly below depth 5.

Depth 10+

Overfitting — train ↑ but val drops to 78–77%.

Max depth

Train acc 91% but val falls to 77.8%.

81.17%

Test Set Accuracy · max_depth = 5

85.2%	80.2%	81.2%
Train Accuracy	**Val Accuracy**	**Test Accuracy**
at depth 5	used for selection	final evaluation

Classification Report (max_depth = 5)

Class	Precision	Recall	F1-score	Support
Not Survived	0.81	0.91	0.86	139
Survived	0.82	0.64	0.72	84
Accuracy			0.81	223
Macro avg	0.81	0.78	0.79	223
Weighted avg	0.81	0.81	0.81	223

Not Survived predicted more accurately (recall 0.91, F1 = 0.86) — majority class advantage.

Macro-avg F1 = 0.79: model performs reasonably on both classes, not just the majority.

81.17% accuracy vs 62% naïve baseline — the model clearly learns meaningful patterns.

Conclusions

👩 Sex is the strongest predictor

Women survived at 74% vs 19% for men — the biggest single feature gap in the dataset.

🚢 Class matters significantly

1st class: 63% survival. 3rd class: 24%. Socioeconomic status had a direct impact on evacuation outcomes.

⚖️ Depth 5 wins— balanced fit

max_depth=5 achieves the best val accuracy (80.24%). Deeper trees overfit from depth 10 onward.

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📈 81.2% test accuracy

A solid baseline for a simple Decision Tree. Room to improve with ensemble methods (Random Forest, Gradient Boosting).

Project Notebook